

SUPPLEMENTARY MATERIAL S1

Candidate readings of the Xiongnu record

Thirty-four readings of Chinese-transcribed Xiongnu names and titles as Turkic words, with the historical parallels that suggested each one. Supplied for the record; not offered as results.

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Supplement to *Reading steppe names: what a transcription channel can and cannot recover from the Chinese record of the Xiongnu*

S1 · author_proposals.csv · compiled 2026

S1.1 What this file is

Section 9 of the paper sets out what a proposed reading of a Xiongnu name has to survive before it can be called a finding: the sense has to be attested in a source independent of the comparison, the morphology has to be licensed by a suffix with the right distribution, and any objection that can be tested has to be tested. The file covers all forty items of the record, and they are not all of one kind. **Two** readings are argued in the paper itself and meet its standard: 若鞮 as *Inakt* (§9) and 冒頓 as *Bögü Tın* (§9.2). **Two** carry a comparison taken from the literature rather than made by us, which the paper then tests: 徑路 as *Kingrak* and 屠耆 as *Doğru* (§9.1). **Two** await a reading, with the frame and the candidate space set out so that one can be proposed. One is a compound pointing at its constituent rows. The remaining **thirty-three** are ours, and none of them meets the standard.

They are here since we produced them over the course of the work. A reader who has seen 稽粥 written out as *kei çuk* will hear Turkish *küçük* whether or not this supplement exists. What can be done is to write them down, date them, label them, and put next to each one the reason it occurred to us, so that the reasoning is inspectable rather than implied.

Nothing in the paper depends on this supplement. The measurements in §§5–8 (the channel accuracies, the period-gap decomposition, the initial-consonant probe, the coda-spelling count) were made without reference to these readings and do not change if every one of them is rejected.

S1.2 Why they are not in the paper

Section 6.3 measures exactly the operation that produced this file, taking a reconstructed reading and looking it up in a Turkic dictionary, against a null model built by resampling Later Han syllables into pseudo-names. Across seven lexicons, from modern Turkish to the *Codex Cumanicus*, between none and three names in thirty-six meet significance of $p \leq 0.05$. At a one-in-twenty threshold, thirty-six coin flips give the same.

Doing the matching by hand does not improve on the algorithm. It makes it worse. A person cannot help preferring the reading that means something, and that preference is precisely the selection effect the null exists to quantify. The algorithm at least fails honestly. A paper that measured the unreliability of this operation and then presented its own outputs as discoveries would be incoherent.

So we keep the two things apart. The paper reports what can be measured. This file records what we imagined while measuring it. Because it is supplementary, it does not carry the paper's evidential standard and should not be read as though it did.

S1.3 How the readings were produced

Every row starts from the same two published inputs, neither of them model output. The Chinese form is from the *Shiji*, *Hanshu* and *Hou Hanshu*. The Later Han reading is Schuessler's. The third column simply respells that reading in Turkish orthography, so that a reader who does not read phonetic notation can see what is being claimed. It adds no information.

The fourth column is ours. We arrive at it by hearing the third column as a Turkic word or compound, usually with a vowel changed, sometimes treating a consonant as lost, and then looking for a historical parallel that would make such a name plausible for a steppe ruler. The fifth column gives the sense that reading would carry. The note beneath each row gives the parallel. Each entry below is set out in two parts: *Phonetics*, which walks the Later Han reading to the proposed name character by character and says which of the steps are ordinary correspondences and which are not; and *Reason*, which gives the word, its attestation where it has one, and the naming parallel that suggested it.

The standard we used for inclusion here is deliberately lower than the paper's, and it is worth stating. A reading is offered when an attested word fits the transcription and nothing in the transcription stands against it. It is not offered when the spelling rules it out, when a character would have to carry a sound the corpora never show it carrying. Under-determination is not a reason to withhold a reading, because almost every short name in the record is under-determined; a demonstrable phonological conflict is a reason.

We had three reasoning habits. We list them because they are the main weakness. The first is the epithet argument: later Turkic rulers are commonly named by a common noun or a natural force: Bayezid I *Yıldırım* "the thunderbolt", Alp Arslan "the lion", Mehmed IV *Avcı* "the hunter", Süleyman I *Kanunî* "the lawgiver", so a reading that yields such a noun is treated as more likely. The second is the career argument: where the Chinese record says what a ruler did, a reading that names the deed is preferred. The third is the compound argument: a long transcription is split into two or three Turkic elements. Each of these makes a reading easier to reach, and none of them makes it more likely to be right. They multiply the search space, which is the opposite of evidence.

S1.4 The two readings argued in the paper

Two readings in this file are argued in the paper rather than merely recorded, because they are supported differently from everything else in it. The first is the recurring element 若鞮 *nök te*, which closes the formal names of six later chanyu, read *Inakt*, "the faithful ones" (§9):

- the *Hou Hanshu* glosses 若鞮 as 孝 "filial", the meaning is recorded in the source, not inferred from a resemblance;
- inak* is attested as *fidelis* "faithful" in the *Codex Cumanicus* (Kuun 1880, p. 182), a medieval source with no connection to the Chinese record;
- the morphology is licensed: *-t* is an Old Turkic plural used with titles (*tegin* → *tegit*, *tarqan* → *tarqat*), and the competing *-tI* reading we ruled out by counting: of 31 Cuman headwords in *-tI*, almost all are past-tense verbs;
- we tested the obvious objection to the spelling and it failed: if the source ended in *-t*, why did the scribe use the open character 鞮? Because 30% of stop-final source syllables in the Sanskrit corpus were written with an open character.

One objection stands and is stated in the paper: if the source word was *inakt*, the initial *l-* is not written, and 若鞮 has no room for it.

The second is 冒頓 *mək tuən^c*, the founder of the confederation, read *Bögü Tın*, "sage spirit" (§9.2). This is where the method yields a positive reading of a Xiongnu name, and it rests on four things rather than a resemblance:

- The frame is tight.** Both codas are written, the *-k* of 冒 and the *-n* of 頓, so both must correspond to something. That fixes four consonants ending in a nasal.
- The first character has a documented Turkic value.** 冒 and 默 share the Later Han reading *mək*, and 默啜 is read *Bögü-Çor*, the personal name of Qapaghan Qaghan. That syllable is on record writing *bögü* in a Turkic ruler's name.
- Every step is separately licensed by a count.** An *m-* character for a source *b-*: 3 of 22 in the Later Han layer. One character covering two source syllables with a stop coda doing the compressing: 47 of the 1,017 verified pairs, as 竭 covers *-gata* in *tathāgata*. And 頓 needs no license at all, since *tın* "spirit, breath" matches with nothing inserted.
- The competing literature reading of 冒頓 fails a test this one passes.** *Bayatur*, the field's answer for a century, ends in *-r* against a written *-n*, a spelling found once in 37 opportunities.

What is unsettled is stated in §9.2 and in the row itself: the compound *bögü tın* is not attested, only its two halves, and the vowel of the second element is not recoverable, so the preference for *tın* over *tün*, *tan* or *tun* is semantic rather than phonological.

Both appear in the tables below in their proper places, so that this file can be read as one list with the standard visible in it.

S1.5 Why the decoder's output is not cited below

The channel of §4 has a decoding step that will return a ranked list of readings for any string in the record. None of those rankings is quoted in this file, and the reason is worth stating, because the numbers look more informative than they are.

A character can be *reading-anchored*, meaning its emission distribution was learned for that character specifically, or it can fall back to a table keyed on the Later Han **initial consonant class alone**, discarding the vowel and the coda. Of the 39 items in the record, only five are reading-anchored throughout; 34 use backoff on at least one character and nine use it on every character. The effect is that whole groups of characters share one distribution:

LATER HAN INITIAL CLASS	CHARACTERS IN THE XIONGNU RECORD SHARING ONE EMISSION DISTRIBUTION	WHAT THE BACKOFF TABLE RETURNS FOR EVERY ONE OF THEM
velar	17: 稽 渠 胡 湖 壺 戶 侯 姑 孤 骨 谷 韓 諧 皋 權 胸 徑	ka 0.34, ga 0.15, ku 0.08
affricate	6: 且 粥 珠 臣 詹 郅	ca 0.18, ja 0.10, cā 0.08
dental stop	9: 鞮 題 當 屠 撐 株 稚 脫 頓	ta 0.23, da 0.11, ti 0.10
liquid	6: 廬 孿 祿 累 蠡 閭	ra 0.37, la 0.17, rī 0.08

TABLE S1 Because 稽 *k^hei^B* and 渠 *giə* draw on the same row, and 粥 *tśuk* and 且 *tso^B* on another, an affricate-plus-velar string and a velar-plus-affricate string return mirror images of one table entry rather than anything specific to those characters. Two further limits apply: the scores are normalised over a retained beam, so they are relative rankings within it and not posteriors; and the syllable inventory is Sanskrit-derived, so *q* and the front rounded vowels *ö* and *ü* cannot be produced at all, so a reading containing *ü* is therefore not rejected by the decoder, it is unavailable to it. §6.2 of the paper makes the general point; this table is its mechanism.

In the entries below, where a spelling question arises, we cite the paper's initial-correspondence counts (§8.1): direct counts of which Chinese initial was used for which source sound across the three corpora. Those are observations rather than model output, and they are what the word "permitted" means wherever it appears in this file.

S1.6 Chanyu, in order of reign

FORM, AS WRITTEN IN THE HISTORIES	LATER HAN READING OF THOSE CHARACTERS (SCHUESSLER)	THE SAME READING IN TURKISH ORTHOGRAPHY	READING WE PROPOSE	SENSE WE PROPOSE FOR THAT READING
頭曼 Touman	do man ^c	do man	Duman	dust cloud
PHONETICS 頭 <i>do</i> → <i>du</i>: the initial is direct; o and u are not separable at the ~50% vowel recovery §7 reports. 曼 <i>man</i> → <i>man</i>: exact, and the coda -n is written, because -n is one of the six codas the Chinese syllable can carry. One vowel change in two syllables. The rival reading <i>tūmen</i> needs <i>ü</i>, which the Later Han script cannot write at all. REASON <i>duman</i> / <i>tuman</i> / <i>touman</i> “smoke, fog, mist” is one word across Kāşgarî, the Codex Cumanicus and modern Turkish. The standard identification is <i>tūmen</i> “ten thousand”, which is absent from the Cuman vocabulary entirely.				
冒頓 Modu / Modun	mək tuən ^c	mık tuın	Bögü Tın	sage spirit
PHONETICS The frame is <i>b/m – k/g – t/d – n</i>, and both codas are written, which makes it one of the tightest in the record. 冒 <i>mək</i> → <i>bögü</i>: three things happen, and each is licensed. The <i>m-</i> stands for a source <i>b-</i>, which in the Later Han corpus happens in 3 of 22 cases, 13.6%. The written -<i>k</i> stands for the source <i>g</i>. And the final vowel of <i>bögü</i> is not written, one character covering two source syllables, a device that occurs in 47 of the 1,017 verified pairs, always with a stop-coda character doing the compressing, as 竭 covers <i>-gata</i> in <i>tathāgata</i> and 弗 covers <i>-putra</i> in <i>śāriputra</i>. There is also a character-level precedent, not merely a rate: 冒 and 默 have the identical Later Han reading <i>mək</i>, and 默噉 is read <i>Bögü-Çor</i>, the personal name of Qapaġhan Qaġhan. That syllable is on record writing <i>bögü</i> in a Turkic ruler's name. 頓 <i>tuən^c</i> → <i>tın</i>: a dental onset with the -<i>n</i> coda written, nothing inserted, nothing discarded. REASON <i>Bögü Tın</i>, “sage spirit”. Kāşgarî glosses <i>bügü</i> as “bilgin, akıllı, hakim”; Clauson says the word “seems to connote both wisdom and mysterious spiritual power”, and it was borrowed early into Mongolian as <i>böge</i>, the ordinary word for a shaman. It appears in regnal styles across the record: <i>Teŋri Bögü Teŋriken</i>, Bögü Qaġhan of the Uyghurs, Bögü-Çor of the Türks. The second element <i>tın</i> is “ruh, nefes” in Kāşgarî and <i>anima</i> in the Codex. Both halves are attested independently of any Chinese source, and the shape, two nouns in apposition, is the commonest Turkic name pattern. Against the standard reading. <i>Bayatur</i> “hero” has been the field's answer for a century. It fails on the point the corpus decides cleanly: it ends in -<i>r</i> against a written -<i>n</i>, a spelling attested in 1 of 37 opportunities, where the scribe's normal choices were an open character or a -<i>t</i>. What is not settled. The compound <i>bögü tın</i> is not itself attested, only its two elements; the vowel of the second is unrecoverable, so <i>tün</i> “night”, <i>tan</i> “dawn” and <i>tun</i> “rest” are equally available and the preference for <i>tın</i> is semantic rather than phonological; and the tone category of 頓 is unused. §9.2 of the paper argues this reading in full.				
稽粥 Jizhou / Jiyu	k ^h ei ^B tśuk	kei çuk	Kiçük	the younger
PHONETICS 稽 <i>k^hei</i> → <i>ki</i>: an aspirated velar is the ordinary vehicle for a source <i>k</i>, and the diphthong <i>ei</i> for <i>i</i> is routine. 粥 <i>tśuk</i> → <i>çük</i>: <i>tś-</i> is the default character class for a source affricate (37% in the Sanskrit corpus), and the coda -<i>k</i> is actually written, the strongest part of the reading, since a stop coda goes unwritten 30% of the time. Only <i>ü</i> is unavailable and surfaces as <i>u</i>. REASON <i>kiçig</i> / <i>kiçik</i> / <i>kiçük</i> “small, younger” is attested three times independently: Kāşgarî has <i>kiçik</i> and <i>kiçük</i> with the derivative <i>kiçiglik</i>, the <i>Kutadgu Bilig</i> has <i>kiçig oġlan erken</i> “when he was a small boy” (1823), and the Codex has <i>Keçak</i> glossed <i>paucus</i>. 稽粥 is the personal name of Laoshang chanyu, Modu's son and successor, so “the younger” fits the succession the Chinese sources record.				

FORM, AS WRITTEN IN THE HISTORIES	LATER HAN READING OF THOSE CHARACTERS (SCHUESSLER)	THE SAME READING IN TURKISH ORTHOGRAPHY	READING WE PROPOSE	SENSE WE PROPOSE FOR THAT READING
軍臣 Junchen	kun džin	kun cin	Künčin	bright as the sun
PHONETICS 軍 <i>kun</i> → <i>kün</i> : initial and coda direct; ü is unwritable and surfaces as u. 臣 <i>džin</i> → <i>čin</i> : dž is the voiced partner of ts, and voicing is leaky in both directions in every corpus measured, so a source ç is reachable through it; the -n coda is written. REASON <i>kün</i> “sun, day”. The Seljuk regnal names Kaykhusraw and Kayqubad carry an association of royal solar radiance through their <i>kay</i> - element.				
詹師廬 Zhanshilu	tšam ši lia	çam ši lia	Çamşılı	dazzling, radiant
PHONETICS 詹 <i>tšam</i> → <i>çam</i> : tš- is the default affricate spelling and the -m coda is written. 師 <i>ši</i> → <i>şı</i> : the retroflex sibilant is the standard vehicle for ş. 廬 <i>lia</i> → <i>lı</i> : note that Chinese l- also carries source r-: in the paper's correspondence counts an r- was written l- in 12 of 14 attestations, so this syllable is genuinely ambiguous between -lı and -rı. REASON The same solar association proposed for <i>Künčin</i> , from a stem of shining or dazzling.				
响犁湖 XuliHu / GouliHu	huo lei ga	huo lei ga	Kalık	sky, heaven
PHONETICS 响 <i>huo</i> → <i>ka</i> , 犁 <i>lei</i> → <i>lı</i> , 湖 <i>ga</i> → <i>k</i> : three onsets, each taken from the majority correspondence, with nothing inserted and no coda discarded. The velar class covers h-, k- and g- alike, so the first and third syllables cost nothing. REASON Kāşgarî glosses <i>kalık</i> “hava, gök, sema” (air, sky, heaven). The word belongs to the same semantic field as 撐犁, the one Xiongnu term the Chinese sources gloss (天 “heaven”), so a ruler bearing it would not be isolated in the record. A second exact fit, <i>kalık</i> “disposition, manner”, is available if the sky reading is thought too convenient.				
伊稚斜 Yizhixie	?i di ^c ja	’i di ya	İdiya	master
PHONETICS 伊 <i>?i</i> → <i>i</i> : a glottal onset is the standard way to write a vowel-initial syllable. 稚 <i>di</i> → <i>di</i> : retroflex ɖ for d is routine. 斜 <i>ja</i> → <i>ya</i> : this character class emits <i>ya</i> more strongly than any other single-syllable correspondence in the record. All three steps are default. REASON <i>idi</i> “lord, master”, attested in Old Turkic. Compare Murad I's epithet <i>Hüdavendigâr</i> , “the sovereign”, a title of the same shape and force.				
烏維 Wuwei	?a wi	’a vi	Abı	one who hunts or stalks
PHONETICS 烏 <i>?a</i> → <i>a</i> : vowel-initial, direct. 維 <i>wi</i> → <i>bı</i> is the weak step: the Chinese character carries a w-, and while Turkic b- and w- alternate, the transcription reads more naturally as <i>avı</i> than <i>abı</i> . This is one of only five items in the record decoded entirely from character-specific evidence, so the mismatch cannot be attributed to a gap in the data. REASON <i>av</i> “hunt”, <i>avcı</i> “hunter”. <i>Avcı</i> names an occupation; <i>abı</i> is proposed as naming a trait. <i>Avcı</i> is also the epithet of Mehmed IV.				
且鞮侯 Qiedihou	tša ^B te go	tša te go	Çatıgka	one ready for the clash
PHONETICS 且 <i>tša</i> → <i>ça</i> : ts- for a source affricate is a minority spelling but a real one, 15% in the Sanskrit corpus and 9% in Ligeti. 鞮 <i>te</i> → <i>tr</i> : direct. 侯 <i>go</i> → <i>gka</i> asks a single Chinese syllable to carry a consonant cluster it structurally cannot, so the reading adds material the transcription does not have. REASON <i>çat</i> - “to strike together, to clash”. Compare Selçuk Bey, whose name has been read as “perpetually ready for the fight”.				
狐鹿姑 Hulugu	ɣua lok ka	ğua lok ka	Kurluga	the encloser, the binder
PHONETICS 姑 <i>ka</i> → <i>ga</i> is well supported by the velar correspondence. The other two syllables are not: 狐 <i>yua</i> → <i>kur</i> supplies an -r that Later Han has no coda slot for, and 鹿 <i>lok</i> → <i>lu</i> discards a coda that was written. Adding an unwritable segment and dropping a written one in the same word pulls in opposite directions. REASON 狐鹿姑 captured the Han general Li Guangli and held the confederation together through a hard decade, so an epithet of binding fits the record. Compare the Göktürk ruler Kapgan Kagan, <i>kapgan</i> “the encloser”.				
壹衍鞮 Huyandi	ga jan ^B te	ga yan te	Kayanti	avalanche
PHONETICS 壹 <i>ga</i> → <i>ka</i> : given a Later Han voiced g-, the source sound is more often k than g, so the devoicing is normal rather than a liberty. 衍 <i>jan</i> → <i>yan</i> : direct, coda written. 鞮 <i>te</i> → <i>tr</i> : direct. All three steps are default or better, phonetically one of the cleaner readings here. REASON <i>kay</i> - “to slide, to slip away”. Parallel in sense to Bayezid I's <i>Yıldırım</i> : a natural force taken as a regnal epithet.				
虛閭權渠 Xulüquanqu	k ^h ia lia gyan gia	kia lia gyan gia	Kalkan	shield

FORM, AS WRITTEN IN THE HISTORIES	LATER HAN READING OF THOSE CHARACTERS (SCHUESSLER)	THE SAME READING IN TURKISH ORTHOGRAPHY	READING WE PROPOSE	SENSE WE PROPOSE FOR THAT READING
PHONETICS 虛 <i>k'ia</i> → <i>ka</i> , 間 <i>lia</i> → <i>l</i> , 權 <i>gyan</i> → <i>kan</i> : the first three characters match exactly, including the written -n coda, at zero minority steps. Only 渠 <i>gia</i> is left unaccounted for, and it may carry a separate title element. REASON <i>kalkan</i> “shield” is in Kâşgarî and in the Codex Cumanicus as <i>Kalkani</i> , “scutum ejus”. A weapon or armour name is an ordinary steppe formation, and it needs no compound to be read.				
握衍胸鞬 Woyanqudi	ʔok jan ^B guo te	’ok yan guo te	Okyan Kutuğtı	blessed sovereign of the arrow flank
PHONETICS 握 ʔok → <i>ok</i> is exact, with the -k coda written, and 衍 <i>jan</i> → <i>yan</i> and 鞬 <i>te</i> → <i>ti</i> are direct. The weak point is 胸 <i>guo</i> → <i>kutuğ</i> , which expands one Chinese syllable into two Turkic ones; nothing in the corpora licenses that expansion. REASON <i>ok</i> “arrow” doubles as a division of the confederation in later Turkic usage (the <i>On Ok</i> , “ten arrows”, of the Western Turks) which is what makes a territorial reading available. <i>kut</i> is “heavenly favour, fortune”.				
呼韓邪 Huhanye	ha gan ja	ha gan ya	Kaganya	kagan of the flank
PHONETICS 呼 <i>ha</i> → <i>ka</i> : an h/k substitution, which the Mongolian corpus supports strongly (a source k or q was written with an h- character 27% of the time). 韓 <i>gan</i> → <i>gan</i> : exact, coda written. 邪 <i>ja</i> → <i>ya</i> : the strongest single correspondence in the record. The written sequence <i>ha gan ya</i> becomes <i>ka-gan-ya</i> on one change. REASON 呼韓邪 commanded the left flank before becoming ruler, so the Chinese record itself supplies the office that the reading names.				
郅支 Zhizhi	tšit tše	çit çe	Çitçi	border-maker, one who carves the boundary
PHONETICS 郅 <i>tšit</i> → <i>çit</i> : tš- is the default affricate vehicle and the -t coda is written. 支 <i>tše</i> → <i>çi</i> : default. Both syllables use the majority correspondence and neither requires an inserted or deleted segment. REASON <i>çit</i> / <i>çiz-</i> , of marking or carving a boundary. 郅支 broke away westward and set up a rival seat, so a name of division fits the career, though it is equally possible that the career suggested the name.				

TABLE S2 The fifteen rulers whose names are not built on the recurring element 若鞬. Reign order follows the *Shiji* and *Hanshu* as reproduced in Appendix A of the paper.

S1.7 The six 若鞬 compounds

FORM, AS WRITTEN IN THE HISTORIES	LATER HAN READING OF THOSE CHARACTERS (SCHUESSLER)	THE SAME READING IN TURKISH ORTHOGRAPHY	READING WE PROPOSE	SENSE WE PROPOSE FOR THAT READING
若鞬 Ruodi	ńak te	nyak te	Inakt	the faithful ones
PHONETICS 若 <i>ńak</i> → <i>inak</i> : a palatal nasal onset with the -k coda written. 鞬 <i>te</i> → <i>-t</i> : the final stop is not written, and Table 7 of the paper licenses that at 30%. One objection stands and the paper states it: if the source was <i>inak</i> , the opening <i>ɪ</i> - is not written and 若鞬 has no room for it. REASON One of the two readings argued in the paper itself, at §9, and it is listed here so this file can be read as a single list with the standard visible in it. <i>Inak</i> is glossed <i>fidelis</i> “faithful” in the Codex Cumanicus; the <i>Hou Hanshu</i> glosses 若鞬 as 孝 “filial”; the plural in <i>-t</i> is an Old Turkic formation used with titles (<i>tegin</i> → <i>tegit</i>); and Clauson records Chagatay <i>inak</i> as “a royal representative or senior minister”, so a court title on this root is what the word ordinarily does.				
復株累若鞬 Fuzhulei Ruodi	buk ʈio lui ńak te	buk tio lui nyak te	Büktel Inakt	Büktel, of the Inakt
PHONETICS 復株累 <i>buk ʈio lui</i> → <i>büktel</i> : the b-, the -k coda, the dental and the liquid all fall in place across the three characters, at zero minority steps, with only the front rounded vowel unwritable. One attested word covers the whole prefix rather than an invented two-element compound. 若鞬 → <i>Inakt</i> is treated in §S1.4. REASON Kâşgarî gives <i>büktel</i> as “orta boylu; yassı arkalı, oturamaklı”: of middling height and flat-backed. A name from a bodily description is a well-attested Turkic type; compare the Hungarian royal name <i>Şişman</i> .				
搜諧若鞬 Souxie Ruodi	so gei ńak te	so gei nyak te	Sarg Inakt	the yellow, the fair, of the Inakt

FORM, AS WRITTEN IN THE HISTORIES	LATER HAN READING OF THOSE CHARACTERS (SCHUESSLER)	THE SAME READING IN TURKISH ORTHOGRAPHY	READING WE PROPOSE	SENSE WE PROPOSE FOR THAT READING
PHONETICS 搜 <i>so</i> → <i>sar</i> : a sibilant onset, which the correspondence counts make obligatory: a source k, q or g was written with a Chinese sibilant in 0 of 2,124 attestations, while Chinese sibilant characters transcribe s, š, č or ś in 98% of 837 cases. The final -r is not written and could not be: Later Han carries no -r in any coda position. 諧 <i>gei</i> → <i>ig</i> : a velar onset, direct. Nothing is inserted, and no character is read as a sound it cannot carry. 若鞮 → <i>Inakt</i> is treated in §S1.4. REASON <i>sarig</i> “sarı, sarı renk” (yellow) is in Kâşgarî, and survives as modern Turkish <i>Sarı</i> , still current both as a name and as an epithet. Color epithets are a documented Turkic naming type: the same dictionary has <i>çakır</i> “grey-eyed”, and <i>Çal</i> “dappled, grey” is proposed for 烏珠留若鞮 on the same footing. Ninety-one attested headwords fit this two-syllable frame, so it does not determine the reading; <i>sarig</i> is offered as the most plausible of them, with nothing in the transcription standing against it.				
車牙若鞮 Cheya Ruodi	tʃʰa ɲa ɳak te	ça nga nyak te	Çöge-Ok Inakt	Arrow of High Eminence, of the Inakt
PHONETICS 車 <i>tʃʰa</i> → <i>çöge</i> : the aspirated affricate is a good vehicle for ç, but ö is unwritable and a second syllable is added. 牙 <i>ɲa</i> → <i>ok</i> : a velar-nasal initial standing for a vowel-initial syllable is unsupported. REASON <i>çöge</i> of height or eminence, with the same <i>ok</i> element.				
烏珠留若鞮 Wuzhuliu Ruodi	ʔa tʃo liu ɳak te	’a ço liu nyak te	Çal Inakt	the grey or dappled one, of the Inakt
PHONETICS 烏 ʔa is vowel-initial, so the reading begins at 珠 <i>tʃo</i> → ç, with 留 <i>liu</i> → l: an affricate written with the default tʃ- character and a liquid, both majority correspondences. Nothing is inserted, and no character is read as a consonant it does not carry. REASON <i>çal</i> is “alaca, kır” in Kâşgarî: dappled, grey. Color epithets are a persistent Turkic naming pattern, as <i>çakır</i> “grey-eyed” shows in the same dictionary.				
烏累若鞮 Wulei Ruodi	ʔa lui ɳak te	’a lui nyak te	Uluğ Inakt	the Great, of the Inakt
PHONETICS 烏 ʔa → u: a glottal-onset character is the standard way to write a vowel-initial syllable. 累 <i>lui</i> → lu: a liquid, direct. Two characters, two segments, nothing supplied that the transcription does not carry. REASON The Codex Cumanicus glosses <i>Oulu</i> as <i>magnus</i> . <i>Uluğ</i> “great” is among the most common elements in Turkic titlature, which is both the strength of this reading and its weakness: a very common word is an easy target to hit.				
呼都而尸道皋若鞮 Huduershī Daogao Ruodi	ha ta nə ʃi dou ⁸ kou ɳak te	ha ta nyɭ ʃi dou kou nyak te	Öz-Uluğ Inakt	Great Core, of the Inakt
PHONETICS Eight characters, and correspondingly the least constrained reading in the file: the longer the string, the more ways there are to split it into Turkic elements, and the split itself is doing most of the work rather than any individual correspondence. 呼 <i>ha</i> → <i>öz</i> and 皋 <i>kou</i> → <i>uluğ</i> are the two largest jumps. REASON <i>öz</i> “self, core” and <i>uluğ</i> “great”, both common Old Turkic elements in titlature.				

TABLE S3 The recurring element itself, then the six later chanyu whose formal names end in it. Only the shared element is argued for in the paper; the material preceding it in each name is a proposal like any other in this file.

s1.8 The ruling clan name, in its two written forms

FORM, AS WRITTEN IN THE HISTORIES	LATER HAN READING OF THOSE CHARACTERS (SCHUESSLER)	THE SAME READING IN TURKISH ORTHOGRAPHY	READING WE PROPOSE	SENSE WE PROPOSE FOR THAT READING
攡鞮 Luandi	lyan te	lyan te	El-indi	state founder
PHONETICS 攡 <i>lyan</i> → <i>el</i> moves the liquid from the onset of the Chinese syllable to the coda of the Turkic one; no corpus shows that kind of metathesis. 鞮 <i>te</i> → <i>indi</i> expands one syllable into two. The reading is driven by the sense rather than by the transcription. REASON <i>el</i> “realm, state” with the definite past in <i>-di</i> : literally “he founded the realm”. This is the clan name as the <i>Shiji</i> and <i>Hanshu</i> write it.				
虛連題 Xulianti	kʰiə lian ⁸ de	kɭa lian de	Kıyalığ El-indi	state founder of the rocky region

FORM, AS WRITTEN IN THE HISTORIES	LATER HAN READING OF THOSE CHARACTERS (SCHUESSLER)	THE SAME READING IN TURKISH ORTHOGRAPHY	READING WE PROPOSE	SENSE WE PROPOSE FOR THAT READING
PHONETICS 虛 <i>k'ia</i> → <i>kɣa</i> is reachable. 連 <i>lian</i> ^B → <i>liŋ</i> adds an unwritten coda, and 題 <i>de</i> → <i>el-indi</i> carries a whole two-syllable word on one character. Three characters are being asked to yield five syllables.				
REASON The same clan name as 攀鞮, in the form the <i>Hou Hanshu</i> gives, with the <i>kaya</i> element recurring from 虛閭權渠.				

TABLE S4 攀鞮 in the *Shiji* and *Hanshu*, 虛連題 in the *Hou Hanshu*. Both are taken here as the same name, which is the usual view.

S1.9 Titles and lexical items

FORM, AS WRITTEN IN THE HISTORIES	LATER HAN READING OF THOSE CHARACTERS (SCHUESSLER)	THE SAME READING IN TURKISH ORTHOGRAPHY	READING WE PROPOSE	SENSE WE PROPOSE FOR THAT READING
單于 chanyu	džan wa	can va	Kan Apa	senior ruler, elder ruler
PHONETICS 單 <i>džan</i> → <i>kan</i> is the weak step: a voiced affricate standing for a velar stop is not a correspondence any of the three corpora show. 于 <i>wa</i> → <i>apa</i> expands one syllable into two. This is one of the five items decoded entirely from character-specific evidence, which makes the mismatch harder to attribute to sparse data.				
REASON <i>apa</i> “elder, ancestor” is attested in Old Turkic titlature. This is the central title of the whole record and the most contested: readings from three language families have been proposed for it.				
撐犁孤塗單于 chengli gutu chanyu	ḍaŋ lei kua da džan wa	dang lei kua da can va	(compound of the three rows below)	
PHONETICS Six characters forming the full ceremonial title, and the only item in the record the Chinese sources gloss in full. It decomposes into three units that each have their own row here: 撐犁 <i>ḍaŋ lei</i> , 孤塗 <i>kua da</i> and 單于 <i>džan wa</i> . No separate reading is proposed for the compound; whatever the three parts are read as, the whole follows.				
REASON The <i>Hanshu</i> glosses the phrase as “Son of Heaven, the Chanyu”, with 撐犁 = 天 “heaven” and 孤塗 = 子 “son”. That makes it the single most valuable item in the record, because both the form and the meaning are recorded, and it is the evidence on which most of the language-attribution argument has rested for a century.				
撐犁 chengli	ḍaŋ lei	dang lei	Tanglı	bearing divine light; sacred
PHONETICS 撐 <i>ḍaŋ</i> → <i>tang</i> : retroflex ɖ for t is routine and the -ŋ coda is written. 犁 <i>lei</i> → <i>lı</i> : direct. Both steps are default. The same two syllables equally support <i>tengri</i> , which the Chinese gloss 天 and the lexicon search of §6.3 both favour: Kāšgarı returns it 4th, the Codex 2nd.				
REASON <i>tang</i> of dawn or light, read as “bearing divine light; sacred”. Offered as an alternative to the standard identification with <i>tengri</i> , not as an improvement on it.				
孤塗 gutū	kua da	kua da	— (awaiting a proposal)	
PHONETICS The frame is kɭg/h – tɖ with the vowels free, and both codas unwritten, so a source -r, -l, -s or -z could hide in either. 66 attested headwords fit it.				
REASON The Chinese sources gloss this one: 子 “son”. That is a rare gift and a hard test. The 66 candidates include <i>kut</i> “heavenly favour, fortune”, <i>kat</i> “layer”, <i>kata</i> “time, occasion”, <i>ketü</i> , <i>Kerti verus</i> , <i>kötü</i> , and not one of them means “son”. The Turkic proposal in the literature is <i>qut</i> “fortune”, which fits the sound and not the gloss; the Yeniseian proposal is (<i>h</i>) <i>ala</i> “son”, which fits the gloss. Appendix A of the paper records this as the strongest single datum on the Yeniseian side.				
屠耆 tuqi	da gɨ	da gɪ	Doğru	straight, upright, honest
PHONETICS 屠 <i>da</i> → <i>do</i> : a dental onset, direct. 耆 <i>gi</i> → <i>ğru</i> : a velar onset, with the -r unwritten, which is forced rather than permitted, since Later Han carries no -r in any coda position. 144 attested headwords fit the frame, so the transcription does little to narrow the choice; the vowels are free as always.				
REASON Not our reading. The Turkic-hypothesis literature reads 屠耆 as <i>*toyri</i> , and the Codex Cumanicus has <i>Togru</i> glossed <i>rectus</i> . The Chinese sources gloss 屠耆 as 賢 “worthy, wise”, and it appears in 左右屠耆王, the Wise Kings of the Left and Right. The senses meet more closely than “straight” suggests. Clauson derives <i>toğuru</i> : / <i>toğru</i> : as a gerund of <i>toğur-</i> , physically “straight” and then metaphorically “straight, honest, upright, true”, and that moral sense is old rather than modern: a fourteenth-century glossary renders it with Arabic <i>al-tiqa</i> “trustworthy, honest”, and <i>toğru ayt-</i> with <i>sadaqa</i> “to tell the truth”. The word survives across the whole Turkic family and as modern Turkish <i>doğru</i> , “straight, correct”. The family also yields names. Kāšgarı records that men are called <i>Toğrı</i> after a bird of prey; an <i>Alp Toğrul Tégin</i> appears in a Uyghur list of proper names; and the first Seljuk sultan bore the name. §9.1 of the paper tests this case alongside 徑路.				

FORM, AS WRITTEN IN THE HISTORIES	LATER HAN READING OF THOSE CHARACTERS (SCHUESSLER)	THE SAME READING IN TURKISH ORTHOGRAPHY	READING WE PROPOSE	SENSE WE PROPOSE FOR THAT READING
谷蠡 luli / guli	kok le ^B	kok le	— (awaiting a proposal)	
PHONETICS The frame is <i>k/g/h – k/g – l/r</i>, with the -k of 谷 written and the coda of 蠡 open. It is unusually tight: only 8 attested headwords fit, against 66 for 孤塗 and 144 for 屠耆. Note that the modern pinyin is read both <i>luli</i> and <i>guli</i>, but the Later Han reading is not in doubt. REASON The eight are <i>kagil</i> “a fresh willow branch used to tie vines”, Codex <i>Kagal lentus</i> “slow, pliant”, Codex <i>Kukel gramen</i> “grass”, <i>Hakil prudentia</i> (an Arabic loan, so excluded on the same grounds as <i>kadi</i>), <i>kafgar</i>, <i>kekre</i>, and two proper names. None is obviously a title. 谷蠡 names a senior office, the Luli Kings ranking below the Wise Kings, so the tightness of the frame makes this a good target: a reading that fits has few competitors.				
當戶 danghu	taŋ ga ^B	tang ga	Tonga	Siberian tiger, leopard
PHONETICS 當 <i>taŋ</i> → <i>ton</i>: direct, with the -ŋ coda written. 戶 <i>go^B</i> → <i>ga</i>: a velar onset, direct. Both characters do work, and the pair is the standard device for writing a medial nasal-plus-velar: in the Sanskrit corpus every source word with a medial <i>ŋg</i> or <i>ŋk</i> is written exactly this way, a -ŋ coda followed by a velar onset: 13 cases of 13, as in 央具梨 for <i>aṅguli</i>, 商羯羅 for <i>śaṅkara</i>, 鶯耆羅 for <i>aṅgiras</i>. Phonetically this is among the least demanding readings in the file: nothing inserted, no coda discarded, no character left over. The vowels are free as always, so <i>Tonga</i>, <i>Tunga</i> and <i>Tanga</i> are equally available. REASON <i>tonga</i> “Siberian tiger, leopard” is the word in the name of the epic hero Alp Er Tonga. Big-cat epithets are a persistent Turkic pattern: Alp Arslan (<i>arslan</i>, lion), Baybars (<i>bars</i>, a large cat). Note that a search of the two lexicons on this frame does not return <i>tonga</i> itself; what it returns is <i>tanuk</i> “witness”, <i>tañuk</i> “a gift presented to a khan” and <i>tüneκ</i> “prison”, of which the second belongs to court vocabulary.				
闕氏 yanzhi / ezhi	ʔat dže ^B	’at ce	Ece	queen
PHONETICS 闕 <i>ʔat</i> → <i>e</i>: a vowel-initial character, but with a -t coda that is written and that the reading discards. 氏 <i>dže^B</i> → <i>ce</i>: direct. Another of the five fully character-anchored items, so the discarded -t is a real cost rather than an artefact. REASON <i>ece</i> “queen”, which survives in modern Turkish with roughly this sense. 闕氏 is the title of the chanyu's principal consort.				
骨都侯 guduhou	kuæt ta go	kuit ta go	Kutak	the blessed one
PHONETICS 骨 <i>kuæt</i> → <i>kut</i>: default velar onset with the -t coda written, the tightest single syllable in this file. 都 <i>ta</i> then begins with the same dental, and Table 8 of the paper measures what that means: at a junction where a written coda is followed by the same-place onset, the source word carries that consonant once rather than twice in 121 of 140 cases (86%). So 骨都 transcribes <i>kuta</i>, not <i>kutta</i>. 侯 <i>go</i> → <i>-k</i> closes it, a Later Han voiced g- standing for a source k more often than for a source g. REASON <i>kut</i> is “kut, uğur, devlet, baht, talih, saadet” in Kāşğarî (heavenly favour, fortune, sovereign grace) and Clauson records the verb <i>kutad-</i> “to be blessed”, the stem behind the <i>Kutadgu Bilig</i>. <i>Kutak</i> itself is not attested: it would be a deverbal or diminutive formation in -(A)k on that stem, which is a live suffix but not one witnessed on this word. Compare <i>Qutalmış</i>, ancestor of the Seljuks of Rum. Note also that <i>kut</i> is the Turkic candidate for 孤塗, where the Chinese gloss 子 “son” fits the Yeniseian proposal better and not this one.				
且渠 juqu	tsa ^B gĩa	tsa gĩa	Çaka	the striker; from çak- ‘to strike fire, to strike a blow’
PHONETICS 且 <i>tsa^B</i> → <i>ça</i>: ts- for a source affricate is a minority spelling but an attested one, 15% in the Sanskrit corpus and 9% in Ligeti, against a tś-character as the default at 37%. 渠 <i>gĩa</i> → <i>ka</i>: a Later Han voiced g- stands for a foreign k more often than for a foreign g. Both steps are permitted by the measured correspondences. REASON <i>çak-</i> “to strike fire from flint, to sting, to slander”, which Clauson describes as onomatopoeic in character and of violent action, physical or mental. No <i>çaka</i> headword exists in Kāşğarî, the Codex or Clauson, only the verb and its derivatives <i>çakmak</i>, <i>çakır</i>, <i>çakılmak</i>, so <i>Çaka</i> would have to be the -A converb of that stem, and a bare converb serving as an office title is not a demonstrable pattern. The 11th-century emir Çaka Bey of Smyrna (Byzantine Τζαχῆς, <i>fl.</i> 1081–93) shows that <i>çak-</i> could yield a personal name, but that is a personal name set against a title. The Chinese sources do not record what the office of 且渠 did, so no sense can be tested against a gloss.				
徑路 jinglu	keŋ ^c la ^c	keng la	Kingrak	a large cleaver-like knife

FORM, AS WRITTEN IN THE HISTORIES	LATER HAN READING OF THOSE CHARACTERS (SCHUESSLER)	THE SAME READING IN TURKISH ORTHOGRAPHY	READING WE PROPOSE	SENSE WE PROPOSE FOR THAT READING
PHONETICS 徑 <i>keŋ</i>^c → <i>king</i>: default velar onset with the -ŋ coda written. 路 <i>la</i>^c → <i>ra</i>: a Chinese l- for a source r-, the direction those counts show in 12 of 14 attestations. The Turkic word's final -k is not written, which Table 7 of the paper licenses at 30%. REASON Not our reading. Friedrich Hirth compared 徑路 with Turkic <i>kıñrak</i> in <i>The Ancient History of China to the End of the Chóu Dynasty</i> (1908), pp. 65–70, calling it the oldest Turkish word on record. Kāšgarī glosses <i>kıñrak</i> as a knife like a butcher's cleaver, for cutting meat and dough, and the Chinese sources describe 徑路 as a blade used in oath-taking, so gloss and referent agree across unconnected sources. Turkic origin, on Clauson's analysis. Clauson derives <i>kıñra:k</i> as a deverbal noun from an unattested <i>*kıñra</i>:-, denominal from <i>kıñır</i>, in the sense of something curved or something that cuts crookedly, so the word is motivated inside Turkic by the standard authority rather than by us. Kāšgarī's index carries the same family (<i>kıñru</i> “askew”, <i>kañrak</i> “palate”, <i>koñur</i>- “to wrench, bend back”), and the Codex glosses <i>Kingir</i> as <i>curvus</i>. Clauson also records the word alive in Teleut as <i>kıñırak</i> and Kirghiz <i>kıñarak</i> / <i>kıñırak</i>, a rough two-edged knife for cutting felt and scraping hides. Two cautions. Kāšgarī's text at III 382 reads <i>kırpa:k</i>, corrected in the margin to <i>kıñra:k</i>, so the form rests on an emendation; and of the family it is <i>kıñır</i> that fits the transcription exactly, while <i>kıñrak</i>, the member meaning a knife, needs the unwritten final -k allowed at 30%. Iranian origin. The object has also been derived from an Iranian word for a short sword, the <i>akinakes</i> type, historically plausible, since the Xiongnu's western neighbors were Iranian-speaking. But 路 is a liquid-onset character, so the channel requires a liquid in second position, and the <i>akinakes</i> frame (k–n–k) has none. As stated, that account fails on the frame. Two cautions there too: we have not seen the exact Iranian etymon its proponents advance, and the channel shows what a transcription permits, never what it requires. §9.1 of the paper sets both origins out.				
服匱 funi	buk ɳik	buk nɪk	Buknik	clay jar (küp)
PHONETICS 服 <i>buk</i> → <i>buk</i>: exact, including the written -k coda. 匱 <i>ɳik</i> → <i>nik</i>: retroflex ɳ for n is routine and the second -k is also written. This is the closest fit to its own transcription of anything in this file: no segment inserted, none discarded. REASON A clay jar (Turkish <i>küp</i>). One of the few Xiongnu material terms the Chinese sources preserve, which is what makes it worth a reading at all.				
甌脫 outuo	ʔo tʰuat	’o tuat	Otuyat	the border watchtowers
PHONETICS 甌 <i>ʔo</i> → <i>o</i>: vowel-initial, direct. 脫 <i>tʰuat</i> → <i>tuyat</i>: the aspirated t- is direct and the -t coda is written; the only addition is the medial -y-. REASON The border watchtowers, or the empty zone between Xiongnu and Han territory, for which the Chinese sources use 甌脫.				
胡祿 hulu	ɡɔ lɔk	ɡo lɔk	Külüg	famous, renowned
PHONETICS 胡 <i>ɡɔ</i> → <i>kü</i>: the devoicing is normal, a Later Han voiced g- standing for a source k more often than for a source ɡ; ü cannot be written and surfaces as o. 祿 <i>lɔk</i> → <i>lüg</i>: the -k coda is written. The consonant frame k–l–k is secure and only the vowels are free. REASON <i>külüg</i> “famous, renowned” is an Old Turkic epithet of praise, recorded by Clauson; note that the form is <i>külüg</i>, not <i>külük</i>, and it is not a headword in either lexicon used here. Two objections stand. First, the Chinese-side record gives the meaning of 胡祿 as “quiver”, not a term of praise, though that gloss is itself marked low-confidence and chronologically shaky. Second, the reading the gloss wants is barred by the transcription: the one Turkic quiver word whose skeleton fits is <i>okluk</i> (Kāšgarī, glossed <i>sadak</i>), and it is vowel-initial, whereas 胡 carries a voiced velar. In the Sanskrit corpus a source vowel-initial syllable was written with a zero or glottal onset in 96% of 183 cases, and with a velar in none. So this is an item where the Chinese gloss names a target, a Turkic word exists for it, and the spelling rules that word out, recorded because a negative of that shape is worth more than a resemblance.				

TABLE S5 The titles and the handful of common nouns the Chinese histories preserve, in the order of Appendix A. The lexical items are the more interesting group because some of them are glossed in the source, which means a proposal about them can in principle be checked.

s1.10 Ethnonyms

FORM, AS WRITTEN IN THE HISTORIES	LATER HAN READING OF THOSE CHARACTERS (SCHUESSLER)	THE SAME READING IN TURKISH ORTHOGRAPHY	READING WE PROPOSE	SENSE WE PROPOSE FOR THAT READING
匈奴 Xiongnu	huoŋ na	huong na	Hung	the ethnonym, from the first character alone
PHONETICS 匈 <i>huoŋ</i> → <i>hung</i>: direct, with the -ŋ coda written. 奴 <i>na</i> is discarded rather than read, on the assumption that 奴 “slave” is a Chinese pejorative appended to the name rather than part of it. REASON Dropping 奴 is a common move in the literature and is not original here; what is proposed is only the vocalisation of the remaining syllable.				
羯 Jie	kiat	kiat	Kıyat	floods

FORM, AS WRITTEN IN THE HISTORIES	LATER HAN READING OF THOSE CHARACTERS (SCHUESSLER)	THE SAME READING IN TURKISH ORTHOGRAPHY	READING WE PROPOSE	SENSE WE PROPOSE FOR THAT READING
PHONETICS A single character. 羯 <i>kiat</i> → <i>kıyat</i> : the k- is default and the -t coda is written, so the consonant frame k...t is secure and only the medial glide is free. Fully character-anchored. REASON <i>kıyat</i> , of flooding. The Jie are recorded as a group within or beside the Xiongnu; the reading is offered for the ethnonym only.				

TABLE S6 Group names rather than personal names, and subject to a further difficulty: a Chinese exonym need not transcribe anything the group called itself.

S1.11 How to disagree

The repository includes `src/lookup_lexicons.py`, which searches Kāşgarî's *Dīwān*, the *Codex Cumanicus* and any EPUB lexicon placed beside them, folding the spelling conventions that separate Turcological from Turkish orthography (q ≡ k, ñ ≡ ng, ş ≡ s, ı ≡ i). Type any proposed form into it and see what the medieval sources actually have. That is how the *Codex* attestation for *inak* was found, and how the -tI count that ruled out the competing suffix was made.

A reader who wants to dispose of a row below should find that easier than we did. The useful form of disagreement is not "this is unlikely", since most of them are, but "this word is not attested before date X", or "this suffix does not attach to that stem", or "the Chinese character cannot carry that coda". Those are the objections that would move something out of this file, in either direction.

S1.12 Status

Most readings in Tables S2–S6 carry the same label in the machine-readable file: *author's proposal; not supported by any measurement in the paper*. That label is the honest summary of them, and it is not weakened by the fact that some of them are more attractive than others. The rows with a different status are named in §S1.1: the two argued in the paper, the two taken from the literature, the two awaiting a reading, and the one compound.

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